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Inventor(s)

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Automatic clinical workflow that recognizes and analyzes 2D and doppler modality echocardiogram images for automated cardiac measurements and diagnosis of cardiac amyloidosis and hypertrophic cardiomyopathy

Abstract

A computer-implemented method for automated diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM) performed by an automated workflow engine executed by at least one processor includes separating a plurality of echocardiogram (echo) images a heart according to 2D images and Doppler modality images. The 2D images are classified by view type, including A4C video. The 2D images are segmented to produce segmented A4C images having a segmentation mask over the left ventricle. Phase detection is performed on the segmented A4C images to determine systole and diastole endpoints per cardiac cycle. Disease classification is performed on beat-to-beat A4C images for respective cardiac cycles. The cardiac cycle probability scores generated for all of the cardiac cycles are aggregated for each A4C video, and the aggregated probability scores for all the A4C videos are combined to generate a patient-level conclusion for CA and HCM.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. in-part of co-pending patent application Ser. No. 17/219,611, filed Mar. 31, 2021, which is a continuation in-part of U.S. patent application Ser. No. 16/216,929, filed on Dec. 11, 2018, which issued on Apr. 28, 2020 as U.S. Pat. No. 10,631,828, the entire disclosures of which are both assigned to the assignee of the present application and incorporated herein by reference.

TECHNICAL FIELD

(1) The disclosed embodiments relate to image and video classification for disease prediction, and more specifically, to an automatic clinical workflow that recognizes and analyzes 2D and Doppler modality echocardiogram images for automated diagnosis of cardiac amyloidosis and hypertrophic cardiomyopathy.

BACKGROUND

(2) Cardiovascular disease including heart failure is a major health problem accounting for about 30% of human deaths worldwide. Heart failure is also the leading cause of hospitalization in adults over the age of 65 years. Echocardiography is an important diagnostic aid in cardiology for the morphological and functional assessment of the heart. In a typical patient echocardiogram (echo) examination, a clinician called a sonographer places an ultrasound device against the patient's chest to capture a number of 2D images/videos of the patients' heart. Reflected sound waves reveal the inner structure of the heart walls and the velocities of blood flows. The ultrasound device position is varied during an echo exam to capture different anatomical sections as 2D slices of the heart from different viewpoints or views. The clinician has the option of adding to these 2D images a waveform captured from various possible modalities including continuous wave Doppler, m-mode, pulsed wave Doppler and pulsed wave tissue Doppler. The 2D images/videos and Doppler modality images are typically saved in DICOM (Digital Imaging and Communications in Medicine) formatted files. Although the type of modality is partially indicated in the metadata of the DICOM file, the ultrasound device position in both the modality and 2D views, which is the final determinant of which cardiac structure has been imaged, is not.

(3) After the patient examination, a clinician/technician goes through the DICOM files, manually annotates heart chambers and structures like the left ventricle (LV) and takes measurements of those structures. The process is reliant on the clinicians' training to recognize the view in each image and make the appropriate measurements. In a follow up examination, a cardiologist reviews the DICOM images and measurements, compares them to memorized guideline values and makes a diagnosis based on the interpretation made from the echocardiogram.

(4) The current workflow process for analyzing DICOM images, measuring cardiac structures in the images and determining, predicting and prognosticating heart disease is highly manual, time-consuming and error prone. Because the workflow process is so labor intensive, more than 95% of the images available in a typical patient echocardiographic study are never annotated or quantified. The view angle or Doppler modality type by which an image was captured is typically not labelled, which means the overwhelming majority of stored DICOMs from past patient studies and clinical trials do not possess the basic structure and necessary identification of labels to allow for machine learning on this data.

(5) There has been a recent proposal for automated cardiac image interpretation to enable low-cost assessment of cardiac function by non-experts. Although the proposed automated system holds the promise of improved performance compared to the manual process, the system has several shortfalls. One shortfall is that the system only recognizes 2D images. In addition, although the proposed system may distinguish between a normal heart and a diseased heart, the proposed system is incapable of distinguishing hearts having similar-looking diseases. Consequently, the number of heart diseases identified by the proposed system is very limited and requires manual intervention to identify other types of heart diseases.

(6) For example, heart failure has been traditionally viewed as a failure of contractile function and left ventricular ejection fraction (LVEF) has been widely used to define systolic function, assess prognosis and select patients for therapeutic interventions. However, it is recognized that heart failure can occur in the presence of normal or near-normal EF: so-called "heart failure with preserved ejection fraction (HFPEF)" which accounts for a substantial proportion of clinical cases of heart failure. Heart failure with severe dilation and/or markedly reduced EF: so-called "heart failure with reduced ejection fraction (HFrEF)" is the best understood type of heart failure in terms of pathophysiology and treatment. The symptoms of heart failure may develop suddenly 'acute heart failure' leading to hospital admission, but they can also develop gradually.

(7) Timely categorization of heart failure subtype-HFrEF or HFPEF and improved risk stratification are critical for the management and treatment of heart failure, but the proposed system does not address this.

(8) The proposed system is also incapable of generating a prognosis based on the identified heart disease and would instead require a cardiologist to manually form the prognosis. The proposed system is further incapable of structuring the automated measurements and labelled views across multiple sources of data, to enable training and validation of disease prediction algorithms across multiple remote patient cohorts.

(9) Transthoracic echocardiography (TTE) is a common imaging modality to screen for Cardiac Amyloidosis (CA), and Hypertrophic Cardiomyopathy (HCM). Multi-parametric evaluation of CA and HCM severity is recommended by current guidelines. There have been other proposals that utilize video-based deep learning 3D CNN models for disease detection on A4C videos: One proposal implements a single model for detection of CA, and a second proposal implements two separate models for detection of CA and HCM, respectively. In particular, the second proposal performs automatic phase detection followed by beat-to-beat analysis of ventricular dimensions on PLAX videos to detect Left Ventricular Hypertrophy (LVH), then used a 3D CNN to detect CA or HCM if LVH is present. The main limitation of these proposals is that they do not include a fully automated pipeline comprising of automated view identification and reporting.

(10) Accordingly, there is a need for an improved and fully automatic clinical workflow that recognizes and analyzes both 2D and Doppler modality echocardiographic images for automated diagnosis of CA and HCM.

BRIEF SUMMARY

(11) The disclosed embodiments provide methods and systems for automated diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM) performed by an automated workflow engine executed by at least one processor. Aspects of the disclosed embodiments include receiving, from a memory, a plurality of echocardiogram images of a heart. The plurality of echocardiogram (echo) images according to 2D images and Doppler modality images. The 2D images are classified by view type, including A4C video. The 2D images are segmented to produce segmented A4C images having a segmentation mask over the left ventricle. Phase detection is performed on the segmented A4C images to determine systole and diastole endpoints per cardiac cycle. Disease classification is performed on beat-to-beat A4C images for the respective cardiac cycles, and gives a set of probabilities for CA, HCM or neither CA or HCM for each cardiac cycle. The cardiac cycle probability scores generated for all of the cardiac cycles are aggregated for each A4C video, and the aggregated probability scores for all the A4C videos are combined to generate a patient-level conclusion for CA and HCM.

(12) According to the method and system disclosed herein, the disclosed embodiments use machine learning to recognize and analyze both 2D and Doppler modality Echocardiographic images for automated measurements and the diagnosis, prediction and prognosis of heart disease, and the system can be deployed in workstation or mobile-based ultrasound point-of-care systems.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

(1) FIGS. 1A-1C are diagrams illustrating embodiments of a system for implementing an automated clinical workflow that recognizes and analyzes both 2D and Doppler modality Echocardiographic images for automated measurements and the diagnosis, prediction and prognosis of heart disease.

(2) FIG. 2 illustrates architectural layers of the echo workflow engine.

(3) FIG. 3A is a flow diagram illustrating one embodiment of a process for performed by the echo workflow engine to automatically recognize and analyze both 2D and Doppler modality echo images for automated measurements and diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM).

- (4) FIG. 3B is a diagram illustrating a video-based disease classification workflow process for automatically detecting CA and HCM.
- (5) FIG. 3C is a diagram illustrating of the video-based disease classification workflow and disease in further detail according to an exemplary embodiment.
- (6) FIG. 3D is a diagram illustrating an example report for CA.
- (7) FIG. 3E is a diagram illustrating an example report for HCM.
- (8) FIG. 4A is a flow diagram illustrating details of the process for automatically recognizing and analyze both 2D and Doppler modality echo images to perform automated measurements and the diagnosis, prediction and prognosis of heart disease according to one embodiment.
- (9) FIG. 4B is a flow diagram illustrating advanced functions of the echo workflow engine.
- (10) FIG. 5A is a diagram illustrating an example 2D echo image.
- (11) FIG. 5B is a diagram illustrating an example Doppler modality image.
- (12) FIGS. 6A-6K are diagrams illustrating some example view types automatically classified by the echo workflow engine.
- (13) FIG. 7 is a diagram illustrating an example 2D image segmented to produce annotations indicating cardiac chambers, and an example Doppler modality image segmented to produce a mask and trace waveform.
- (14) FIGS. 8A and 8B are diagrams illustrating examples of finding systolic/diastolic end points.
- (15) FIGS. 9A and 9B are diagrams illustrating processing of an imaging window in a 2D echo image and the automated detection of out of sector annotations.
- (16) FIGS. 10A and 10B are diagrams graphically illustrating structural measurements automatically generated from annotations of cardiac chambers in a 2D image, and velocity measurements automatically generated from annotations of waveforms in a Doppler modality.
- (17) FIG. 11 is a diagram graphically illustrating measurements of global longitudinal strain that were automatically generated from the annotations of cardiac chambers in 2D images.
- (18) FIG. 12A is a diagram graphically illustrating an example set of best measurement data based on largest volume cardiac chambers and the saving of the best measurement data to a repository.
- (19) FIG. 12B is a diagram graphically illustrating the input of automatically derived measurements from a patient with normal LV EF measurements into a set of rules to determine a conclusion that the patient has normal diastolic function, diastolic dysfunction, or indeterminate.
- (20) FIG. 13 is a diagram graphically illustrating the output of classification, annotation and measurement data to an example JSON file.
- (21) FIG. 14 is a diagram illustrating a portion of an example report showing highlighting values that are outside the range of international guidelines.
- (22) FIG. 15 is a diagram illustrating a portion of an example report of Main Findings that may be printed and/or displayed by the user.
- (23) FIG. 16A is a diagram showing a graph A plotting PCWP and HFpEF scores.
- (24) FIG. 16B is a diagram showing a graph B plotting CV mortality or hospitalization for HF.

DETAILED DESCRIPTION

(25) The disclosed embodiments relate to an automatic clinical workflow that recognizes and analyzes 2D and Doppler modality Echocardiographic images and videos for automated measurements diagnosis of cardiac amyloidosis and hypertrophic cardiomyopathy. The following description is presented to enable one of ordinary skill in the art to make and use the invention and is provided in the context of a patent application and its requirements. Various modifications to the disclosed embodiments and the generic principles and features described herein will be readily apparent. The disclosed embodiments are mainly described in terms of particular methods and systems provided in particular implementations. However, the methods and systems will operate effectively in other implementations. Phrases such as “disclosed embodiment”, “one embodiment” and “another embodiment” may refer to the same or different embodiments. The embodiments will be described with respect to systems and/or devices having certain components. However, the

systems and/or devices may include more or less components than those shown, and variations in the arrangement and type of the components may be made without departing from the scope of the invention. The disclosed embodiments will also be described in the context of particular methods having certain steps. However, the method and system operate effectively for other methods having different and/or additional steps and steps in different orders that are not inconsistent with the disclosed embodiments. Thus, the present invention is not intended to be limited to the embodiments shown, but is to be accorded the widest scope consistent with the principles and features described herein.

(26) The disclosed embodiments provide method and system for implementing a software-based automatic clinical workflow using machine learning that recognizes and analyzes both 2D and Doppler modality Echocardiographic images for automated measurements and diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM), and which can be deployed in workstation or mobile-based ultrasound point-of-care systems.

(27) The disclosed workflow includes automated identification of 2D videos, such as A4C, for disease classification. The workflow's disease classification model performs beat-to-beat detection of both CA and HCM, allowing a machine learning (ML) model to implicitly study diastolic and systolic functions in reference with CA and HCM. In addition, a multi-beat algorithm references multiple cardiac cycles whenever possible, to generate a more confident final diagnosis of CA/HCM or no CA/HCM.

(28) FIGS. 1A-1C are diagrams illustrating embodiments of a system for implementing an automated clinical workflow that recognizes and analyzes both 2D and Doppler modality Echocardiographic images for automated measurements and the diagnosis, prediction and prognosis of heart disease. FIG. 1A shows a basic standalone configuration for the automated clinical workflow system **10A** and a connected configuration **10B**. The automated clinical workflow **10A** is primarily implemented as a software application, referred to as echo workflow engine **12**, that executes on a computer **14** operating in a standalone setting, disconnected from other devices on network **26**. The computer **14** may be implemented in any form factor including a workstation, desktop, notebook, laptop server or tablet capable of running an operating system, such as Microsoft Windows® (e.g., Windows 7®, Windows 10®), Apple macOS®, Linux®, Apple iOS®, Android®, and the like.

(29) The computer **14** may include typical hardware components (not shown) including a processor, input devices (e.g., keyboard, pointing device, microphone for voice commands, buttons, touchscreen, etc.), output devices (e.g., a display device, speakers, and the like), and wired or wireless network communication interfaces (not shown) for communication. The computer **14** may include internal computer-readable media, such as memory (not shown) containing computer instructions comprising the echo workflow engine **12**, which implements the functionality disclosed herein when executed by one or more computer processors.

(30) The computer **14** may further include local internal storage for storing one or more databases **16** and an image file archive **18**. In one embodiment, the contents of the image file archive **18** include echocardiogram image files (also referred to herein as echo images), which in some embodiments may be stored in DICOM (Digital Imaging and Communications in Medicine) format.

(31) In one embodiment, the computer **14** is in communication with peripheral devices such a point-of-care (POC) device **25**, an ultrasound imaging device **24**, or both. The POC device **25** is capable of measuring cardiac biomarkers in POC environments such as an emergency room, intensive care unit, physician's office, an ambulance, a patient setting, and remote emergency sites.

(32) In one embodiment, the computer **14** is in communication with peripheral devices such an ultrasound imaging device **24** that captures echocardiogram images of a patient's organ (e.g., a heart), which may then be stored as a patient study using the database **16** and image file archive **18**. For example, the computer **14** may be located in a hospital or clinical lab environment where

Echocardiography is performed as a diagnostic aid in cardiology for the morphological and functional assessment of the heart. During a typical patient echocardiogram exam (referred to as a study), a sonographer or technician places the ultrasound imaging device **24** against the patient's chest to capture 2D echo images/videos of the heart to help diagnose the particular heart ailment. Measurements of the structure and blood flows are typically made using 2D slices of the heart and the position of the ultrasound imaging device **24** is varied during an echo exam to capture different anatomical sections of the heart from different viewpoints. The technician has the option of adding to these 2D echo images a waveform captured from various possible modalities including: continuous wave Doppler, m-mode, pulsed wave Doppler and pulsed wave tissue Doppler. The 2D images and Doppler waveform images may be saved as DICOM files. Although the type of modality is sometimes indicated in the metadata of the DICOM file, the 2D view is not.

(33) The computer **14** may further include removable storage devices such as an optical disk **20** and/or a flash memory **22** and the like for storage of the echo images. In some embodiments, the removable storage devices may be used as an import source of echo images and related data structures into the internal image file archive **18**, rather than or in addition to, the ultrasound imaging device **24**. The removable storage devices may also be used as an archive for echocardiogram data stored in the database **16** and/or the image file archive **18**.

(34) FIG. **1A** also shows an advanced optional embodiment, referred to as connected configuration **10B**, where the computer **14** may be connected through the network **26** and a router **28** to other DICOM based devices, such as DICOM servers **30**, network file share devices **32**, echo workstations **34**, and/or cloud storage services **36** hosting DICOM files. In the connected configuration **10B**, several possible interactions with the database **16** and the image file archive **18** are possible, as described below.

(35) One possible interaction is to use the cloud storage services **36** as an internal archive. In case of very large archives consisting of large amounts of DICOM files, the computer **14** may not have sufficient storage to host all files and the echo workflow engine **12** may be configured to use external network storage of the cloud storage services **36** for file storage.

(36) Another possible interaction is to use the cloud storage services **36** as an import source by i) selecting a DICOM file set or patient study, which includes the DICOM and Doppler waveforms images and patient data and examination information. The patient study may also be selected by a reserved DICOMDIR file instance, from which the patient, exams and image files are read.

(37) Yet a further possible interaction is to use the DICOM servers **30**, the network file share devices **32**, echo workstations **34**, and/or DICOM clients (of FIG. **1C**) acting as DICOM servers (workstations with modalities CFind, CMove and CStore) in order to retrieve patients, exams and images by performing a CFind operation, followed by a CMove operation, to request the remote device to send the images resulting from the CFind operation.

(38) Referring now to FIG. **1B**, a handheld configuration **10C** for the automated clinical workflow system is shown. In this embodiment, the computer **14** of FIG. **1A** is implemented as a handheld device **14'**, such as a tablet or a mobile phone, connected to a wired or wireless portable ultrasound scanner probe **24'** that transmits echo images to the handheld device **14'**. In such embodiments, the echo workflow engine **12** may be implemented as an application executed by the handheld device **14'**.

(39) FIG. **1C** illustrates a software as a service (SaaS) configuration **10D** for the automated clinical workflow. In this embodiment, the echo workflow engine **12** is run on a server **40** that is in communication over the network **26** with a plurality of client devices **42**. In this embodiment, the server **40** and the echo workflow engine **12** may be part of a third-party service that provides automated measurements and the diagnosis, prediction and prognosis of heart disease to client devices (e.g., hospital, clinic, or doctor computers) over the Internet. It should be understood that although the server **40** is shown as a single computer, it should be understood that the functions of server **40** may be distributed over more than one server. In an alternative embodiment, the server **40**

and the echo workflow engine **12** of FIG. **1C** may be implemented as a virtual entity whose functions are distributed over multiple client devices **42**. Likewise, it should be understood that although the echo workflow engine **12** is shown as a single component in each embodiment, the functionality of the echo workflow engine **12** may be separated into a greater number of modules/components.

(40) Conventionally, after a patient examination where echo images are captured stored, a clinician/technician goes through the DICOM files, manually annotates heart chambers and structures and takes measurements, which are presented in a report. In a follow up examination, a doctor will review the DICOM images and measurements, compare them to memorized guideline values and make a diagnosis. Such a process is reliant on the clinicians' training to recognize the view and make the appropriate measurements so that a proper diagnosis can be made. Such a process is error-prone and time consuming.

(41) According to the disclosed embodiments, the echo workflow engine **12** mimics the standard clinical practice, processing DICOM files of a patient by using a combination of machine learning, image processing, and DICOM workflow techniques to derive clinical measurements, diagnose specific diseases, and prognosticate patient outcomes, as described below. While an automated solution to echo image interpretation using machine learning has been previously proposed, the solution only analyzes 2D echo images and not Doppler modality waveform images. The solution also mentions disease prediction, but only attempts to handle two diseases (hypertrophic cardiomyopathy and cardiac amyloidosis) and the control only compares normal patients to diseased patients.

(42) The echo workflow engine **12** of the disclosed embodiments, however, improves on the automated solution by utilizing machine learning to automatically recognize and analyze not only 2D echo images but also Doppler modality waveform images. The echo workflow engine **12** is also capable of comparing patients having similar-looking heart diseases (rather than comparing normal patients to diseased patients), and automatically identifies additional diseases, including both heart failure with reduced ejection fraction (HFrEF) and heart failure with preserved ejection fraction (HFpEF). HFrEF is known as heart failure due to left ventricular systolic dysfunction or systolic heart failure and occurs when the ejection fraction is less than 40%. HFpEF is a form of congestive heart failure where the amount of blood pumped from the heart's left ventricle with each beat (ejection fraction) is greater than 50%. Finally, unlike the proposed automated solution, the echo workflow engine **12** automatically generates a report with the results of the analysis for medical decision support, detection of CA and HCM.

(43) Cardiac amyloidosis is a rare, progressive disease characterized by the accumulation of abnormal proteins called amyloids in the heart tissue. These amyloid deposits can disrupt the normal structure and function of the heart, leading to various symptoms and complications. Hypertrophic cardiomyopathy is a genetic heart condition characterized by abnormal thickening (hypertrophy) of the heart muscle, primarily affecting the left ventricle. This thickening can make it harder for the heart to pump blood effectively, leading to various symptoms and complications.

(44) Traditional echocardiography to evaluate CA and HCM is highly manual, time consuming, error-prone, limited to specialists, and involves long waiting times. However, the Artificial Intelligence (AI) approached described herein allows fully automated, fast and reproducible echocardiographic image analysis; turning a manual process of 30 minutes, 250 clicks, with 21% variability, into an AI-automated process taking 2 minutes, 1 click, with 0% variability. Such AI-enabled echocardiographic interpretation therefore not only increases efficiency and accuracy, but also opens the door to decision support for non-specialists.

(45) FIG. **2** illustrates architectural layers of the echo workflow engine **12**. In one embodiment, the echo workflow engine **12** may include a number of software components such as software servers or services that are packaged together in one software application. For example, the echo workflow engine **12** may include a machine learning layer **200**, a presentation layer **202**, and a database layer

204.

(46) The machine learning layer **200** comprises several neural networks to process incoming echo images and corresponding metadata. The neural networks used in the machine learning layer may comprise a mixture of different classes or model types. In one embodiment, machine learning layer **200** utilizes a first set of one or more neural networks **200A** to classify 2D images by view type, and uses a second set of neural networks **200B** to both extract features from Doppler modality images and to use the extracted features to classify the Doppler modality images by region (the neural networks used to extract features may be different than the neural network used to classify the images). The first set of neural networks **200A** and the second set of neural networks **200B** may be implemented using convolutional neural network (CNN) and may be referred to as classification neural networks or CNNs.

(47) Additionally, a third set of neural networks **200C**, including adversarial networks, are employed for each classified 2D view type in order to segment the cardiac chambers in the 2D images and produce segmented 2D images. A fourth set of neural networks **200D** are used for each classified Doppler modality region in order to segment the Doppler modality images to generate waveform traces. In additional embodiments, the machine learning layer **200** may further include a set of one or more prediction CNNs for disease prediction and optionally a set of one or more prognosis CNNs for disease prognosis (not shown). The third and fourth sets of neural networks **200C** and **200D** may be implemented using CNNs and may be referred to as segmentation neural networks or CNNs.

(48) In machine learning, a CNN is a class of deep, feed-forward artificial neural network typically used for analyzing visual imagery. Each CNN comprises an input and an output layer, as well as multiple hidden layers. In neural networks, each node or neuron receives an input from some number of locations in the previous layer. Each neuron computes an output value by applying some function to the input values coming from the previous layer. The function that is applied to the input values is specified by a vector of weights and a bias (typically real numbers). Learning in a neural network progresses by making incremental adjustments to the biases and weights. The vector of weights and the bias are called a filter and represents some feature of the input (e.g., a particular shape).

(49) The machine learning layer **200** operates in a training mode to train each of the CNNs **200A-200D** prior to the echo workflow engine **12** being placed in an analysis mode to automatically recognize and analyze echo images in patient studies. In one embodiment, the CNNs **200A-200D** may be trained to recognize and segment the various echo image views using thousands of echo images from an online public or private echocardiogram DICOM database. In one embodiment, the CNNs are trained for both automated view identification and) automated annotation of relevant CA and HCM views in the echo images. A CNN may be additionally trained for automatic phase detection and beat-to-beat direct classification of disease from the relevant view. Beat-to-beat direct classification is the classification of disease when given only the video duration from one cardiac cycle. The beat-to-beat disease classifications are aggregated to produce a classification for the overall video, and the disease classifications for all the videos in a patient study are combined to produce a final disease classification.

(50) The presentation layer **202** is used to format and present information to a user. In one embodiment, the presentation layer is written in HTML 5, Angular 4 and/or JavaScript. The presentation layer **202** may include a Windows Presentation Foundation (WPF) graphical subsystem **202A** for implementing a lightweight browser-based user interface that displays reports and allows a user (e.g., doctor/technician) to edit the reports. The presentation layer **202** may also include an image viewer **202B** (e.g., a DICOM viewer) for viewing echo images, and a python server **202C** for running the CNN algorithms and generating a file of the results in JavaScript Object Notation (JSON) format, for example.

(51) The database layer **204** in one embodiment comprises a SQL database **204A** and other external

services that the system may use. The SQL database **204A** stores patient study information for individual patient studies input to the system. In some embodiments, the database layer **204** may also include the image file archive **18** of FIG. **1A**.

(52) FIG. **3A** is a flow diagram illustrating one embodiment of a process for performed by the echo workflow engine **12** to automatically recognize and analyze both 2D and Doppler modality echo images for automated measurements and diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM). For evaluation of CA and HCM, the 2D image view type used is apical 4-chamber (A4C). The process occurs once the echo workflow engine **12** is trained and placed in analysis mode.

(53) The process may begin by the echo workflow engine **12** receiving from a memory a plurality of echocardiogram images taken by an ultrasound device of a heart (block **300**). In one embodiment, the patient study may include 70-90 images and videos.

(54) A first module of the echo workflow engine **12** may be used to operate as a filter to separate the plurality of echocardiogram images according to 2D images and Doppler modality images based on analyzing image metadata (block **302**). The first module analyzes the DICOM tags, or metadata, incorporated in the image, and runs an algorithm based upon the tag information to distinguish between 2D and modality images, and then separate the modality images into either pulse wave, continuous wave, PWTDI or m-mode groupings. A second module of the echo workflow engine **12** may perform color flow analysis on extracted pixel data using a combination of analyzing both DICOM tags/metadata and color content within the images, to separate views that contain color from those that do not. A third module may anonymize the data by removing metatags that contain personal information and cropping the images to exclude any identifying information. A fourth module may extract the pixel data from the images and converts the pixel data to numpy arrays for further processing. These functions may be performed by a lesser number or a greater number of models in other embodiments.

(55) Because sonographers do not label the view types in the echo images, one or more of the neural networks are used to classify the echo images by view type. In one embodiment, the first set of one or more neural networks is used by the echo workflow engine **12** to classify the 2D images by view type (block **304**); and the second set of neural networks is used by the echo workflow engine **12** to classify the Doppler modality images by Doppler modality region (block **306**). As shown, the processing of 2D images is separate from the processing of Doppler modality images. In one embodiment, the first and second neural networks may be implemented using the set of classification convolutional neural networks (CNNs) **200A**.

(56) In one specific embodiment, a five class CNN may be used to classify the 2D images by view type and an 11 class CNN may be used to classify the Doppler modality images by Doppler modality region. In an alternative embodiment, the first and second neural networks may be combined into one neural network. In one embodiment, the classification of the echo images by view type and regions by the neural networks can be based on a majority voting scheme to determine the optimal answer. For example, a video can be divided into still image frames, and the workflow engine generates for each image frame a classification label, where the label constitutes a vote, and the classification label receiving the highest number of the votes is applied as the classification of the video.

(57) In one embodiment, the echo workflow engine **12** is trained to classify many different view types. For example, the echo workflow engine **12** may be configured to classify any number of different view types including: parasternal long axis (PLAX), apical 2-, 3-, and 4-chamber (A2C, A3C, and A4C), A4C plus pulse wave of the mitral valve, A4C plus pulse wave tissue Doppler on the septal side, A4C plus pulse wave tissue Doppler on the lateral side, A4C plus pulse wave tissue Doppler on the tricuspid side, A5C plus continuous wave of the aortic valve, A4C+Mmode (TrV), A5C, and by different regions such as: CW AoV (Continuous Wave Aortic Valve Doppler) and PW LVOT (Pulsed Wave Left Ventricular Outflow Tract Doppler).

(58) Based on the classified images, a third set of neural networks is used by the echo workflow engine **12** to segment regions of interest (e.g., cardiac chambers) in the 2D images to produce annotated or segmented 2D images, including apical 4-chamber (A4C) video images (block **308**). In one embodiment, a neural network trained on videos is used to analyze the A4C images to detect CA and HCM.

(59) The process of segmentation includes determining locations where each of the cardiac chambers begin and end to generate outlines of structures of the heart (e.g., cardiac chambers) depicted in each image and/or video. FIG. **6D** is a diagram illustrating an example A4C view type, and FIG. **7** illustrates a diagram illustrating the A4C view segmented to produce segments or annotations **700**.

(60) A fourth set of neural networks is used by the echo workflow engine **12** on each classified Doppler modality region to generate waveform traces for the Doppler modality images to generate annotated or segmented Doppler modality images (block **309**). The process of segmentation produces a trace of an outline of the waveform depicting the velocity of blood flow in a Doppler modality image.

(61) In one embodiment, the third and fourth sets of neural networks may be referred to as segmentation neural networks and may comprise the set of segmentation CNNs **200B** and **200C**. The choice of segmentation CNN used is determined by the view type of the image, which makes the prior correct classification of view type a crucial step. In a further embodiment, once regions of interest are segmented, a separate neural network can be used to smooth outlines of the segmentations.

(62) In one embodiment, the segmentation CNNs may be trained from hand-labeled real images or artificial images generated by general adversarial networks (GANs). One or more segmentation CNNs may be trained for semantic segmentation of the CW AoV and PW LVOT images. From the segmentation of each wave, relevant measurements can be calculated.

(63) Using both the segmented 2D images and the segmented Doppler modality images, the echo workflow engine **12** calculates measurements of cardiac features of the heart for predetermined types of diseases (block **310**).

(64) The echo workflow engine **12** then generates a conclusion for a disease outcome, including by comparing the calculated measurements of cardiac features to cardiac guidelines for the predetermined types of diseases (block **312**).

(65) The echo workflow engine **12** further outputs at least one report to a user showing disease outcomes, including for the predetermined types of diseases, ones of the calculated measurements that fall within or outside of the guidelines (block **314**). In one embodiment, two reports are generated and output: the first report is a list of the calculated values for each measurement with the highest confidence as determined by a rules based engine, highlighting values among the calculated measurements that fall outside of the International guidelines; and the second report is a comprehensive list of all measurements calculated on every image frame of every video, in every view, generating large volumes of data.

(66) All report data and extracted pixel data may be stored in a structured database to enable machine learning and predictive analytics on images that previously lacked the quantification and labelling necessary for such analysis. The structured database may be exported to a cloud-based server or may remain on premises (e.g., of the lab owning the images) and can be connected to remotely. By connecting these data sources into a single network, the disclosed embodiments can progressively train disease prediction algorithms across multiple network nodes and validate the algorithms in distinct patient cohorts. In one embodiment, the reports may be electronically displayed to a doctor and/or a patient on a display of an electronic device and/or as a paper report. In some embodiments, the electronic reports may be editable by the user per rule or role-based permissions, e.g., a cardiologist may be allowed to modify the report, but a patient may have only view privileges.

(67) FIG. 3B is a diagram illustrating a video-based disease classification workflow process for automatically detecting CA and HCA. The process may begin by segmenting, with a 2D CNN, the left ventricle of the heart in the image frames of an A4C video to produce segmented A4C images having a segmentation mask over the left ventricle (block 320—corresponds to FIG. 3A, block 308). A 2D-CNN is a standard convolution neural network generally used on image data.

(68) Automatic phase detection is performed on the segmented A4C images by a phase detection algorithm to determine systole endpoints and diastole endpoints per cardiac cycle in the A4C video, wherein the images frames of the A4C video between the systole and diastole endpoints comprise beat-to-beat video images (block 322). In one embodiment, the automatic phase detection may include performing shape analysis on the segmentation mask in the segmented A4C images to estimate systole and diastole durations of the video.

(69) FIG. 3C is a diagram illustrating of the video-based disease classification workflow and disease in further detail according to an exemplary embodiment. A4C videos 350 are shown input to 2D CNN 352, the output of which is used to determine systole and diastole endpoints 354 per cardiac cycle.

(70) Referring again to FIG. 3B, CA/HCM disease classification is performed by a 3D CNN, which receives the beat-to-beat images for respective cardiac cycles defined by pairs of systole endpoints and diastole endpoints, and outputs cardiac cycle probability scores per the respective cardiac cycles for at least one of three cardiac cycle outcomes: i) Cardiac Amyloidosis (CA), ii) Hypertrophic Cardiomyopathy (HCM), and iii) No CA or HCM (block 324). A 3D CNN uses a three-dimensional filter to perform convolutions.

(71) In one embodiment, the 3D CNN model is trained to learn spatiotemporal patterns in A4C videos to identify CA, HCM or neither. Specifically to do so, when given a cardiac cycle video, the model outputs a vector containing the three probability scores. In one embodiment, the 3D CNN for CA/HCM disease classification may output probability scores between 0-1 for each cardiac cycle outcome. Prior to computing the probability scores, the workflow may filter out the A4C video images having low probability scores.

(72) Referring again to FIG. 3C, beat-to-beat video images 356 of the A4C video images for a single cardiac cycle are shown input to 3D CNN 358, which outputs a probability or prediction per cardiac cycle for CA, HCM or neither CA or HCM.

(73) Referring again to FIG. 3B, a multi-beat algorithm aggregates the cardiac cycle probability scores generated for all of the cardiac cycles to generate a A4C video-level conclusion for at least one of three disease outcomes: i) Cardiac Amyloidosis (CA), ii) Hypertrophic Cardiomyopathy (HCM), and iii) No CA or HCM (block 326). The workflow may filter out the A4C video-level disease predictions having low probability scores.

(74) Responsive to a patient study including a plurality of A4C videos, the multi-beat algorithm 362 may further combine the A4C video-level conclusions from the plurality of A4C videos to generate a patient-level conclusion for the at least one of three disease outcomes (block 328).

(75) Referring again to FIG. 3C, cardiac cycle probability scores 360 are shown input to the multi-beat algorithm 362, which aggregates the cardiac cycle probability scores from all the cycles and all the A4C videos and outputs a patient-level CA/HCM conclusion 364. Finally, a report is output to the user showing the patient-level conclusion for the at least one of three disease outcomes (330).

(76) FIG. 3D is a diagram illustrating an example report for CA, and FIG. 3E is a diagram illustrating an example report for HCM.

(77) FIG. 4A is a flow diagram illustrating further details of the process for automatically recognizing and analyze both 2D and Doppler modality echo images to perform automated measurements and the diagnosis, prediction and prognosis of heart disease according to one embodiment.

(78) The process may begin with receiving one or more patient studies (FIG. 3 block 300), which

comprises blocks **400-4010**. In one embodiment, echo images from each of the patient studies are automatically downloaded into the image file archive **18** (block **400**). The echo images may be received from a directly from local or remote storage source of the computer **14**. The local storage sources may include internal/external storage of the computer **14** including removable storage devices. The remote storage sources may include the ultrasound imaging device **24**, the DICOM servers **30**, the network file share devices **32**, the echo workstations **34**, and/or the cloud storage services **36** (see FIG. **1A**). In one embodiment, the echo workflow engine **12** includes functions for operating as a picture archiving and communication server (PACS), which is capable of handling images from multiple modalities (source machine types, one of which is the ultrasound imaging device **24**). The echo workflow engine **12** uses PACS to download and store the echo images into the image file archive **18** and provides the echo workflow engine **12** with access to the echo images during the automated workflow. The format for PACS image storage and transfer is DICOM (Digital Imaging and Communications in Medicine).

(79) Patient information from each of the patient studies is extracted and stored in the database **16** (block **402**). Non-image patient data may include metadata embedded within the DICOM images and/or scanned documents, which may be incorporated using consumer industry standard formats such as PDF (Portable Document Format), once encapsulated in DICOM. In one embodiment, received patient studies are placed in a processing queue for future processing, and during the processing of each patient study, the echo workflow engine **12** queues and checks for unprocessed echo images (block **404**). The echo workflow engine **12** monitors the status of patient studies and keeps track of them in a queue to determine which have been processed and which are still pending. In one embodiment, prioritization of the patient studies in the queue may be configured by a user. For example, the patient studies may be prioritized in the queue for processing according to the date of the echo exam, the time of receipt of the patient study or by estimated severity of the patient's heart disease.

(80) Any unprocessed echo images are then filtered for having a valid DICOM image format and non DICOM files in an echo study are discarded (block **406**). In one embodiment, the echo images are filtered for having a particular type of format, for example, a valid DICOM file format, and any other file formats may be ignored. Filtering the echo images for having a valid image file format enhances the reliability of the echo workflow engine **12** by rejecting invalid DICOM images for processing.

(81) Any unprocessed valid echo images are then opened and processed in the memory of the computer **14** (block **408**). Opening of the echo images for the patient study in memory of the computer **14** is done to enhance processing speed by echo workflow engine **12**. This is in contrast to an approach of opening the echo files as sub-processes, saving the echo files to disk, and then reopening each echo image during processing, which could significantly slow processing speed.

(82) The echo workflow engine **12** then extracts and stores the metadata from the echo images and then anonymizes the images by blacking out the images and overwriting the metadata in order to protect patient data privacy by covering personal information written on the image (block **410**). As an example, DICOM formatted image files include metadata referred to as DICOM tags that may be used to store a wide variety of information such as patient information, Doctor information, ultrasound manufacture information, study information, and so on. In one embodiment, the extracted metadata may be stored in the database **16** and the metadata in image files is overwritten for privacy.

(83) After receipt and processing of the patient studies, the echo workflow engine **12** separates 2D images from Doppler modality images so the two different image types can be processed by different pipeline flows, described below. In one embodiment, the separating of the images (FIG. **3**, block **302**) may comprise blocks **412-414**. First, the 2D images are separated from the Doppler modality images by analyzing the metadata (block **412**).

(84) FIGS. **5A** and **5B** are diagrams illustrating an example 2D echo image **500** and an example

Doppler modality image **502** including a waveform, respectively. The echo workflow engine **12** may determine the image type by examining metadata/DICOM tags. In one embodiment, information within the DICOM tags may be extracted in order to group the images into one of the following four classes: 2D only, pulsed-wave (PW), continuous wave (CW), and m-mode. Similarly, the transducer frequency of the ultrasound imaging device **24** in the metadata may be used to further separate some of the PW images into a fifth class: pulsed-wave tissue doppler imaging (PWTDI).

(85) Referring again to FIG. **4A**, the echo workflow engine **12** may also filter out images with a zoomed view, which may also be determined by analyzing the metadata (block **414**). Any of the echo images that has been zoomed during image capture are not processed through the pipeline because when zooming, useful information is necessarily left out of the image, meaning the original image would have to be referenced for the missing data, which is a duplication of effort that slows processing speed. Accordingly, rather than potentially slowing the process in such a manner, the echo workflow engine **12** filters out or discards the zoomed images to save processing time. In an alternative embodiment, filtering out zoomed images in block **414** may be performed prior to separating the images in block **412**.

(86) After separating the 2D images from the Doppler modality images, the echo workflow engine **12** extracts and converts the image data from each echo image into numerical arrays (block **416**). For the echo images that are 2D only, the pixel data comprises a series of image frames played in sequence to create a video. Because the image frames are unlabeled, the view angle needs to be determined. For the Doppler modality images that include waveform modalities, there are two images in the DICOM file that may be used for subsequent view identification, a waveform image and an echo image of the heart. The pixel data is extracted from the DICOM file and tags in the DICOM file determine the coordinates to crop the images. The cropped pixel data is stored in numerical arrays for further processing. In one embodiment, blocks **412**, **414** and **416** may correspond to the separating images block **302** of FIG. **3A**.

(87) After separating images, the echo workflow engine **12** attempts to classify each of the echo images by view type. In one embodiment, view classification (FIG. **3** blocks **304** and **306**) corresponds to blocks **418-422**.

(88) According to the disclosed embodiments, the echo workflow engine **12** attempts to classify each of the echo images by view type by utilizing parallel pipeline flows. The parallel pipeline includes a 2D image pipeline and a Doppler modality image pipeline. The 2D pipeline flow begins by classifying, by a first CNN, the 2D images by view type (block **418**), corresponding to block **304** from FIG. **3A**. The Doppler modality image pipeline flow begins by classifying, by a second CNN, the Doppler modality images by view type (block **420**), corresponding to block **306** from FIG. **3A**.

(89) FIGS. **6A-6K** are diagrams illustrating some example view types automatically classified by the echo workflow engine **12**. As stated previously, example view types may include parasternal long axis (PLAX), apical 2-, 3-, and 4-chamber (A2C, A3C, and A4C), A4C plus pulse wave of the mitral valve, A4C plus pulse wave tissue Doppler on the septal side, A4C plus pulse wave tissue Doppler on the lateral side, A4C plus pulse wave tissue Doppler on the tricuspid side, A5C plus continuous wave of the aortic valve, A4C+Mmode (TrV), A5C+PW (LVOT).

(90) Referring again to FIG. **4A**, in one embodiment, 2D image classification is performed as follows. If the DICOM file contains video frames from a 2D view, only a small subset of the video frames is analyzed to determine 2D view classification for more efficient processing. In one embodiment, the subset of the video frames may range approximately 8-12 video frames, but preferably 10 frames are input into one of the CNNs **200A** trained for 2D to determine the actual view. In an alternative embodiment, a subset of video frames may be randomly selected from the video file. In one embodiment, the CNNs **200A** classify each of the analyzed video frames as one of: A2C, A3C, A4C, A5C, PLAX Modified, PLAX, PSAX AoV level, PSAX Mid-level, Subcostal

Ao, Subcostal Hep vein, Subcostal IVC, Subcostal LAX, Subcostal SAX, Suprasternal and Other. (91) Doppler modality images comprise two images, an echocardiogram image of the heart and a corresponding waveform, both of which are extracted from the echo file for image processing. In one embodiment, Doppler modality image classification of continuous wave (CW), pulsed-wave (PW), and M-mode images is performed as follows. If the DICOM file contains a waveform modality (CW, PW, PWTDI, M-mode), the two extracted images are input to one of the CNNs **200A** trained for CW, PW, PWTDI and M-mode view classification to further classify the echo images as one of: CW (AoV), CW (TrV), CW Other, PW (LVOT), PW (MV), PW Other, PWTDI (lateral), PWTDI (septal), PWTDI (tricuspid), M-mode (TrV) and M-mode Other.

(92) There are many more potential classifications available for modalities, but the present embodiments strategically select the classes above, while grouping the remaining potential classes into “Other”, in order to maximize processing efficiency, while identifying the most clinically important images for further processing and quantification. Customization of the CNNs **200A** occurs in the desired number of layers used and the quantity of filters within each layer. During the training phase, the correct size of the CNNs may be determined through repeated training and adjustments until optimal performance levels are reached.

(93) During view classification, the echo workflow engine **12** maintains classification confidence scores that indicate a confidence level that the view classifications are correct. The echo workflow engine **12** filters out the echo images having classification confidence scores that fail to meet a threshold, i.e., low classification confidence scores (block **422**). Multiple algorithms may be employed to derive classification confidence scores depending upon the view in question. Anomalies detected in cardiac structure annotations, image quality, cardiac cycles detected, and the presence of image artifacts may all serve to decrease the classification confidence score and discard an echo image out of the automated echo workflow.

(94) With respect to the confidence scores, the echo workflow engine **12** generates and analyzes several different types of confidence scores at different stages of processing, including classification, annotation, and measurements (e.g., blocks **422**, **434** and **442**). For example, poor quality annotations or classifications, which may be due to substandard image quality, are filtered out by filtering the classification confidence scores. In another example, in a patient study the same view may be acquired more than once, in which case the best measurements are chosen by filtering out low measurement confidence scores as described further below in block **442**. Any data having a confidence score that meets a predetermined threshold continues through the workflow. Should there be a duplication of measurements both with high confidence, the most clinically relevant measurement may be chosen.

(95) Next the echo workflow engine **12** performs image segmentation to define regions of interest (ROI). In computer vision, image segmentation is the process of partitioning a digital image into multiple segments (sets of pixels) to locate and boundaries (lines, curves, and the like) of objects. Typically, annotations are a series of boundary lines overlaying overlaid on the image to highlight segment boundaries/edges. In one embodiment, the segmentation to define ROI (FIG. 3 block **308**) corresponds to blocks **426-436**.

(96) In one embodiment, the 2D image pipeline annotates, by a third CNN, regions of interests, such as cardiac chambers in the 2D images, to produce annotated 2D images (block **426**). An annotation post process then erodes the annotations to reduce their dimensions, spline fits outlines of cardiac structures and adjusts locations of the boundary lines closer to the region of interest (ROIs) (block **427**). The 2D image pipeline continues with analyzing the ROIs (e.g., cardiac chambers) in the annotated 2D images to estimate volumes and determine key points in the cardiac cycle by finding systolic/diastolic end points (block **430**). For 2D only views, measurements are taken at the systolic or diastolic phase of the cardiac cycle, i.e. when the left ventricle reaches the smallest volume (systole) or the largest volume (diastole). From the 2D video images, it must be determined which end points are systolic and which are diastolic based on the size of the estimated

volumes of the left ventricle. For example, a significantly large left ventricle may indicate a dystonic end point, while a significantly small volume may indicate a systolic end point. Every video frame is annotated, and the volume of the left ventricle is calculated throughout the whole cardiac cycle. The frames with minimum and maximum volumes are detected with a peak detection algorithm.

(97) The Doppler modality pipeline analyzes the Doppler modality images and generates, by a fourth CNN, a mask and a waveform trace in the Doppler modality images to produce annotated Doppler modality images (block **431**).

(98) FIG. **7** is a diagram illustrating an example 2D image segmented in block **426** to produce annotations **700** indicating cardiac chambers, and an example Doppler modality image segmented in block **431** to produce a mask and waveform trace **702**.

(99) In one embodiment, the third and fourth CNNs may correspond to segmentation CNNs **200B**. In one embodiment, each of the CNNs **200B** used to segment the 2D images and Doppler modality images may be implemented as U-Net CNN, which is convolutional neural network developed for biomedical image segmentation. Multiple U-Nets may be used. For example, for 2D images, a first U-Net CNN can be trained to annotate ventricles and atria of the heart from the A2C, A3C, A4C views. A second U-net CNN can be trained to annotate the chambers in the PLAX views. For M-mode views, a third U-Net CNN can be trained to segment the waveform, remove small pieces of the segments to find likely candidates for the region of interest, and then reconnect the segments to provide a full trace of the movement of the mitral valve. For CW views, a fourth U-net CNN can be trained to annotate and trace blood flow. For PW views, a fifth U-net CNN trained to annotate and trace the blood flow. For PWTDI views, a sixth U-net CNN can be trained to annotate and trace movement of the tissues structures (lateral/septal/tricuspid valve).

(100) Referring again to FIG. **4A**, the Doppler modality pipeline continues by processing the annotated Doppler modality images with a sliding window to identify cycles, peaks are measured in the cycles, and key points in the cardiac cycle are determined by finding systolic/diastolic end points (block **432**). Typically, a Doppler modality video may capture three heart cycles and the sliding window is adjusted in size to block out two of the cycles so that only one selected cycle is analyzed. Within the selected cycle, the sliding window is used to identify cycles, peaks are measured in the cycles, and key points in the cardiac cycle are determined by finding systolic/diastolic end points.

(101) FIGS. **8A** and **8B** are diagrams illustrating examples of finding systolic/diastolic end points in the cardiac cycle, for both 2D and Doppler modalities, respectively, which are key points in order to take accurate cardiac measurements.

(102) Referring again to FIG. **4A**, in one embodiment, the echo workflow engine **12** maintains annotation confidence scores corresponding to the estimated volumes, systolic/diastolic end points, identified cycles and measured peaks. The echo workflow engine **12** filters out annotated images having annotation confidence scores that fail to meet a threshold, i.e., low annotated confidence scores (block **434**). Examples of low confidence annotations may include annotated images having one or more of: excessively small areas/volumes, sudden width changes, out of proportion sectors, partial heart cycles, and insufficient heart cycles.

(103) After images having low annotation confidence scores are filtered out, the echo workflow engine **12** defines an imaging window for each image and filters out annotations that lie outside of the imaging window (block **435**).

(104) FIGS. **9A** and **9B** are diagrams illustrating processing of an imaging window in a 2D echo image **900**. In FIG. **9A**, a shaded ROI **902** is shown having unshaded regions or holes **904** therein. Open CV morphology transformations are used to fill the holes **904** inside the ROI, as shown in FIG. **9B**. Thereafter, a Hough line transformation may be used to find an imaging window border **906**. As is well known, a Hough transform is a feature extraction technique used in digital image processing to find imperfect instances of objects within a certain class of shapes. After an imaging

window border is found, a pixel account of annotations beyond the imaging window border is made. Annotations **908** with a significant number of pixels outside the border of the imaging window are then discarded.

(105) Referring again to FIG. **4A**, in the handheld configuration for the automated clinical workflow system **10C** (See FIG. **1B**), the patient studies may not include Doppler modality images. According to a further aspect, the disclosed embodiments accommodate for such handheld configurations by using the 2D images to simulate Doppler modality measurements by using Left Ventricular (LV) and Left Atrial (LA) volume measurements to derive E, e' and A (e.g., early and late diastolic transmittal flow and early/mean diastolic tissue velocity) measurements (block **436**). In one embodiment, simulating the Doppler modality measurements may be optional and may be invoked based on a software setting indicating the presence of a handheld configuration and/or absence of Doppler modality images for the current patient study.

(106) Referring again to FIG. **4A**, in one embodiment, once cardiac features are annotated during segmentation, the cardiac features are then measured during a measurement process (block **310** of FIG. **3A**), which in one embodiment may comprise block **438-448**. The process of measuring cardiac features may begin by quantifying a plurality of measurements using the annotations. First, the 2D pipeline measures the 2D images left/right ventricle, left/right atriums, left ventricular outflow (LVOT) and pericardium (block **438**). For the Doppler modality images, the Doppler modality image pipeline measures blood flow velocities (block **440**).

(107) More specifically, for A2C, A3C, A4C, and A5C image views, volumetric measurements of chamber size are conducted on the systolic and diastolic frames of the video, and image processing techniques mimic a trained clinician at measuring the volume using the method of disks (MOD). For 2D Plax, PSAX (mid-level), PSAX (AoV level), Subcostal, Suprasternal and IVC image views, linear measurements of chamber size and inter-chamber distances are conducted on the systolic and diastolic frames of the video using image processing techniques to mimic the trained clinician. For M-mode image views, from the annotated segments of the movement of the Tricuspid Valve, a center line is extracted and smoothed, and then the peaks and valleys are measured in order to determine the minimum and maximum deviations over the cardiac cycle. For PW image views, from the annotations of the blood flow, a mask is created to isolate parts of the waveform. A sliding window is then run across the trace to identify one full heart cycle, in combination with heart rate data from the DICOM tags, to use as a template. This template is then used to identify all other heart cycles in the image. Peak detection is then performed on each cycle and then run through an algorithm to identify which part of the heart cycle each peak represents. For CW image views, from the annotations of the trace of the blood flow, curve fitting is performed on the annotation to then quantify the desired measurements. For PWTDI image views, from the annotations of the movement of the tissue, a mask is created to isolate parts of the waveform. A sliding window is then run across the trace to identify one full heart cycle, in combination with heart rate data from the DICOM tags, to use as a template. This template is then used to identify all other heart cycles in the image. Peak detection is then performed on each cycle and then run through an algorithm to identify which part of the heart cycle each peak represents.

(108) FIGS. **10A** and **10B** are diagrams graphically illustrating structural measurements automatically generated from annotations of cardiac chambers in a 2D image, and velocity measurements automatically generated from annotations of waveforms in a Doppler modality, respectively.

(109) The echo workflow engine **12** is configured to produce many measurements. Referring again to FIG. **4A**, in one embodiment, the echo workflow engine **12** maintains measurement confidence scores corresponding to the measured left/right ventricles, left/right atriums, LVOTs, pericardium and measured velocities. The echo workflow engine **12** filters out echo images having measurement confidence scores that fail to meet a threshold, i.e., low measurement confidence scores (block **442**).

(110) Measurement of cardiac features continues with calculating longitudinal strain graphs using the annotations generated by the CNNs (block **444**). Thereafter, a fifth CNN is optionally used to detect pericardial effusion **446**.

(111) FIG. **11** is a diagram graphically illustrating measurements of global longitudinal strain that were automatically generated from the annotations of cardiac chambers in 2D images.

(112) Referring again to FIG. **4A**, for all remaining non-filtered out data, the echo workflow engine **12** selects as best measurement data the measurements associated with cardiac chambers with the largest volumes, and saves with the best measurement data, image location, classification, annotation and other measurement data associated with the best measurements (block **447**).

(113) FIG. **12A** is a diagram graphically illustrating an example set of best measurement data **1200** based on largest volume cardiac chambers and the saving of the best measurement data **1200** to a repository, such as database **16** of FIG. **1A**.

(114) Referring again to FIG. **4A**, the echo workflow engine **12** then generates conclusions by inputting the best measurement data **1200** to a set of rules based on international measurement guidelines to generate conclusions for medical decisions support (block **448**).

(115) FIG. **12B** is a diagram graphically illustrating the input of normal LV EF measurements into a set of rules to determine a conclusion that the patient has normal diastolic function, diastolic dysfunction, or indeterminate.

(116) Referring again to FIG. **4A**, after the conclusions are generated, a report is generated and output (FIG. **3** block **314**), which may comprise blocks **450-456**. Report generation may begin by the echo workflow engine **12** outputting the best measurement data **1200** to a JSON file for flexibility of export to other applications (block **450**).

(117) FIG. **13** is a diagram graphically illustrating the output of classification, annotation and measurement data to an example JSON file.

(118) Referring again to FIG. **4A**, a lightweight browser-based user interface (UI) is displayed showing a report that visualizes the best measurement data **1200** from the JSON file and that is editable by a user (e.g., doctor/technician) for human verification (block **452**). As is well known, a lightweight web browser is a web browser that is optimized to reduce consumption of system resources, particularly to minimize memory footprint, and by sacrificing some of the features of a mainstream web browser. In one embodiment, any edits made to the data are stored in the database **16** and displayed in the UI.

(119) In order to make clinically relevant suggestion to the user, measurements associated with the best measurement data **1200** are automatically compared to current international guideline values and any out-of-range values are highlighted for the user (block **454**).

(120) FIG. **14** is a diagram illustrating a portion of an example report showing highlighting values that are outside the range of international guidelines.

(121) Referring again to FIG. **4A**, the user is provided with an option of generating a printable report showing an automated summary of Main Findings (i.e., a conclusion reached after examination) and underlining measurements of the patient's health (block **456**).

(122) FIG. **15** is a diagram illustrating a portion of an example report of Main Findings that may be printed and/or displayed by the user.

(123) In one embodiment, the automated workflow of the echo workflow engine **12** may end at block **456**. However, in further aspects of the disclosed embodiments, the process may continue with advance functions, as described below.

(124) FIG. **4B** is a flow diagram illustrating advanced functions of the echo workflow engine. According to this embodiment, the echo workflow engine **12**, takes as inputs values of specific measurements that were automatically derived using machine learning (see block **310**), and analyzes the specific measurements to determine disease diagnosis/prediction/prognosis versus both disease and matched controls and normal patient controls (block **460**). In one embodiment, the echo workflow engine **12** may use the disease predictions to perform diagnosis of any combination

of: cardiac amyloidosis, hypertrophic cardiomyopathy, restrictive pericarditis, cardiotoxicity, early diastolic dysfunction and Doppler free diastolic dysfunction assessment (block **462**). A prognosis in the form of an automated score may then be generated to predict mortality and hospitalizations in the future (block **464**).

(125) Echocardiography is key for the diagnosis of heart failure with preserved ejection fraction (HFpEF). However, existing guidelines are mixed in their recommendations for echocardiogram criteria and none of the available guidelines have been validated against gold-standard invasive hemodynamic measurements in HFpEF.

(126) According to one embodiment, the echo workflow engine **12** further generates a diagnostic score for understanding predictions (block **466**). Using machine learning, the echo workflow engine **12** validates the diagnostic score against invasively measured pulmonary capillary wedge pressure (PCWP) and determines the prognostic utility of the score in a large HFpEF cohort.

(127) In one embodiment, the echo workflow engine **12**, takes as the inputs values of specific measurements that were automatically derived using machine learning workflow, and analyzes the input values using an HFpEF algorithm to compute the HFpEF diagnostic score.

(128) Recognizing that hypertensive heart disease is the most common precursor to HFpEF and has overlapping echocardiogram characteristics with HFpEF, echocardiogram features of 233 patients with HFpEF (LVEF $\geq$ 50%) was compared to 273 hypertensive controls with normal ejection fraction but no heart failure. An agnostic model was developed using penalized logistic regression model and Classification and Regression Tree (CART) analysis. The association of the derived echocardiogram score with invasively measured PCWP was investigated in a separate cohort of 96 patients. The association of the score with the combined clinical outcomes of cardiovascular mortality of HF hospitalization was investigated in 653 patients with HFpEF from the Americas echocardiogram sub study of the TOPCAT trial.

(129) According to one embodiment, left ventricular ejection fraction (LVEF $<$ 60%), peak TR velocity ($>$ 2.3 m/s), relative wall thickness (RWT $>$ 0.39 mm), interventricular septal thickness ($>$ 12.2 mm) and E wave ($>$ 1 m/s) are selected as the most parsimonious combination of variables to identify HFpEF from hypertensive controls. A weighted score (range 0-9) based on these 5 echocardiogram variables had a combined area under the curve of 0.9 for identifying HFpEF from hypertensive controls.

(130) FIGS. **16A** and **16B** are diagrams showing a graph A plotting PCWP and HFpEF scores, and a graph B plotting CV mortality or hospitalization for HF, respectively. Graph A shows that in the independent cohort, the HFpEF score was significantly associated with PCWP in patients with HFpEF (R^{sup.2}=0.22, P=0.034). Graph B shows that a one-point increase was associated with a 12% increase in risk (hazard ratio [HR] 1.12; 95% CI 1.02-1.23, P=0.015) for the combined outcome after multivariable correction.

(131) According to the disclosed embodiments, the echocardiographic score can distinguish HFpEF from hypertensive controls and is associated with objective measurements of severity and outcomes in HFpEF.

(132) A method and system for implementing a software-based automatic clinical workflow using machine learning that recognizes and analyzes both 2D and Doppler modality echocardiographic images for automated measurements and diagnosis of cardiac amyloidosis and hypertrophic cardiomyopathy, and which can be deployed in workstation or mobile-based ultrasound point-of-care systems has been disclosed. The present disclosure has been described in accordance with the embodiments shown, and there could be variations to the embodiments, and any variations would be within the spirit and scope of the present invention. For example, the disclosed embodiments can be implemented using hardware, software, a computer readable medium containing program instructions, or a combination thereof. Accordingly, many modifications may be made by one of ordinary skill in the art without departing from the spirit and scope of the appended claims.

Claims

1. A computer-implemented method for automated diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM) performed by an automated workflow engine executed by at least one processor, the method comprising: receiving, from a memory, a plurality of echocardiogram images of a heart; separating the plurality of echocardiogram (echo) images according to 2D images and Doppler modality images; classifying the 2D images by view type, including apical 4-chamber (A4C) video; segmenting, by a 2D convolutional neural networks (CNN), a left ventricle of the heart in image frames of the A4C video to produce segmented A4C images having a segmentation mask over the left ventricle; performing automatic phase detection on the segmented A4C images to determine systole endpoints and diastole endpoints per cardiac cycle in the A4C video, wherein the image frames of the A4C video between the systole and diastole endpoints comprise beat-to-beat video images; performing, by a 3D CNN, CA and HCM disease classification by receiving the beat-to-beat images for respective cardiac cycles, and outputting cardiac cycle probability scores per the respective cardiac cycles for at least one of three cardiac cycle outcomes: i) CA, ii) HCM, and iii) No CA or HCM; aggregating the cardiac cycle probability scores generated for the respective cardiac cycles to generate a A4C video-level conclusion for at least one of three disease outcomes: i) CA, ii) HCM, and iii) No CA or HCM; responsive to a patient study including a plurality of A4C videos, combining A4C video-level conclusions from the plurality of A4C videos to generate a patient-level conclusion for the at least one of three disease outcomes; and outputting a report showing the patient-level conclusion for the at least one of three disease outcomes.
2. The method of claim 1, wherein receiving, from a memory, the plurality of echo images, further comprises: receiving, by the processor, the plurality of echo images directly from a local or remote source, including an ultrasound device; storing the plurality of echo images in an image archive; and opening the stored echo images in the memory for processing.
3. The method of claim 1, wherein separating the echo images further comprises: analyzing metadata incorporated in the echo images to distinguish between the 2D and the Doppler modality images; separating the Doppler modality images into either pulse wave, continuous wave, PWTDI or m-mode groupings; performing color flow analysis on extracted pixel data using a combination of the metadata and color content within the echo images to separate views that contain color from those that do not; removing from the echo images any metatags that contain personal information and cropping the echo images to exclude any identifying information; and extracting pixel data from the echo images and converting the pixel data to numpy arrays for further processing.
4. The method of claim 1, wherein classifying the 2D images and the Doppler modality images is based on a majority voting scheme comprising: dividing a video of a 2D image of a Doppler modality image into frames; generating for the frames, classification labels that constitute votes; and applying the classification label receiving a highest number of the votes as the classification of the video.
5. The method of claim 1, further comprising: implementing the workflow engine to comprise a first set of one or more classification CNNs for view classification, a second set of one or more segmentation CNNs for chamber segmentation and waveform mask/trace, a third set of one or more prediction CNNs for disease prediction.
6. The method of claim 5, wherein the one or more segmentation CNNs are trained from hand-labeled real images or artificial images generated by general adversarial networks (GANs).
7. The method of claim 1, further comprising: for all non-filtered out data, selecting as best measurement data the measurements associated with cardiac chambers with largest volumes; and saving with the best measurement data, image location, classification, annotation and other measurement data associated with the best measurement data.

8. A system, comprising: a memory storing a plurality of echocardiogram images taken by an ultrasound device of a heart; at least one processor coupled to the memory; and a workflow engine, which when executed by the at least one processor is configurable to: receive, from the memory, a plurality of echocardiogram images of the heart; separate the plurality of echocardiogram (echo) images according to 2D images and Doppler modality images; classify the 2D images by view type, including apical 4-chamber (A4C) video; segment, by a 2D CNN, a left ventricle of the heart in image frames of the A4C video to produce segmented A4C images having a segmentation mask over the left ventricle; perform automatic phase detection on the segmented A4C images to determine systole endpoints and diastole endpoints per cardiac cycle in the A4C video, wherein the images frames of the A4C video between the systole and diastole endpoints comprise beat-to-beat video images; perform, by a 3D CNN, CA and HCM disease classification by receiving the beat-to-beat images for respective cardiac cycles, and outputting cardiac cycle probability scores per the respective cardiac cycles for at least one of three cardiac cycle outcomes: i) CA, ii) HCM, and iii) No CA or HCM; aggregate the cardiac cycle probability scores generated for the respective cardiac cycles to generate a A4C video-level conclusion for at least one of three disease outcomes: i) CA, ii) HCM, and iii) No CA or HCM; responsive to a patient study including a plurality of A4C videos, combine the A4C video-level conclusions from the plurality of A4C videos to generate a patient-level conclusion for the at least one of three disease outcomes; and output a report showing the patient-level conclusion for at least one of three disease outcomes.

9. The system of claim 8, wherein the workflow engine receives from the plurality of echo images directly from a local or remote source, including an ultrasound device; stores the plurality of echo images in an image archive; and opens the stored echo images in the memory for processing.

10. The system of claim 8, wherein the workflow engine separates the echo images uses metadata incorporated in the echo images to distinguish between the 2D and the Doppler modality images; separates the Doppler modality images into either pulse wave, continuous wave, PWTDI or m-mode groupings; performs color flow analysis on extracted pixel data using a combination of the metadata and color content within the echo images to separate views that contain color from those that do not; removes from the echo images any metatags that contain personal information and cropping the echo images to exclude any identifying information; and extracts pixel data from the echo images and converts the pixel data to numpy arrays for further processing.

11. The system of claim 8, wherein the workflow engine classifies the 2D images and the Doppler modality images based on a majority voting scheme, wherein workflow engine divides a video of a 2D image of a Doppler modality image into frames; generates for the frames, classification labels that constitute votes; and applies the classification label receiving a highest number of the votes as the classification of the video.

12. The system of claim 8, wherein the workflow engine is implemented to comprise a first set of one or more classification CNNs for view classification, a second set of one or more segmentation CNNs for chamber segmentation and waveform mask/trace, a third set of one or more prediction CNNs for disease prediction.

13. The system of claim 12, wherein the one or more segmentation CNNs are trained from hand-labeled real images or artificial images generated by general adversarial networks (GANs).

14. An executable software product stored on a non-transitory computer-readable medium containing program instructions for implementing an automated workflow for diagnosis of cardiac amyloidosis (CA) and hypertrophic cardiomyopathy (HCM), which when executed by a set of one or more processors, are configurable to cause the set of one processors to perform operations comprising: receiving, from a memory, a patient study comprising a plurality of echocardiogram images taken by an ultrasound device of a heart; separating, by a filter, the plurality of echocardiogram (echo) images according to 2D images and Doppler modality images based on analyzing image metadata; classifying the 2D images by view type, including apical 4-chamber (A4C) video; segmenting, by a 2D CNN, a left ventricle of the heart in image frames of the A4C

video to produce segmented A4C images having a segmentation mask over the left ventricle; performing automatic phase detection on the segmented A4C images to determine systole endpoints and diastole endpoints per cardiac cycle in the A4C video, wherein the images frames of the A4C video between the systole and diastole endpoints comprise beat-to-beat video images; performing, by a 3D CNN, CA and HCM disease classification by receiving the beat-to-beat images for respective cardiac cycles, and outputting cardiac cycle probability scores per the respective cardiac cycles for at least one of three cardiac cycle outcomes: i) CA, ii) HCM, and iii) No CA or HCM; aggregating the cardiac cycle probability scores generated for the cardiac cycles to generate a A4C video-level conclusion for at least one of three disease outcomes: i) CA, ii) HCM, and iii) No CA or HCM; responsive to a patient study including a plurality of A4C videos, combining the A4C video-level conclusions from the plurality of A4C videos to generate a patient-level conclusion for the at least one of three disease outcomes; and outputting a report showing the patient-level conclusion for at least one of three disease outcomes.
